

Internship Project Documentation

PROJECT REPORT

Crop Insights: A Tableau-Based Analysis of India's Agricultural Crop Production

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Project Name	Indian Agriculture Crop Production Analysis
Dataset Period	1997 - 2021
Technology Stack	Python, Tableau, Scikit-learn, Plotly
Repository	github.com/ram-1709/Indian-s-Agriculture-Crop-Production-Analysis

1. Introduction

1.1 Project Overview

Crop Insights is a Tableau-based data analytics project that examines a historical agricultural crop production dataset for India, spanning 1997 to 2021, covering crop production, cultivated area, season, state, district, and yield across the country. The project builds an interactive dashboard and guided storyboard to help users compare crop production by state, crop type, season, and year, supported by a Python-based machine learning layer for production forecasting.

1.2 Purpose

The purpose of this project is to give farmers, policymakers, researchers, and agribusiness analysts a centralized, visual way to compare agricultural performance across states, crops, and seasons — replacing scattered, static historical records with an interactive dashboard and data-backed, forward-looking insights.

2. Ideation Phase

2.1 Problem Statement

Farmers, policymakers, and agribusiness analysts lack a centralized, visual way to compare crop production across states, crops, seasons, and years. Historical agricultural records are spread across static tables with no aggregated view, making it difficult to identify high-performing states, seasonal patterns, or emerging production risks without manual, time-consuming analysis (full Customer Problem Statement table to be documented in the Ideation phase of the project repository).

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A policy analyst reviewing state agricultural performance	compare crop production across states and years without manual spreadsheet work	historical records are static and not visually comparable	there is no single, centralized, visual tool covering multi-decade, multi-state crop data	overwhelmed and slowed down, relying on manual cross-referencing instead of data
PS-2	A farmer or agribusiness planner choosing which crop/season to prioritize	understand which crops and seasons yield the best production in their region	seasonal and regional patterns are buried in raw, unaggregated rows	state and season-level trends are not visually surfaced anywhere	unable to confidently plan cultivation or resource allocation

2.2 Empathy Map Canvas

An empathy map was built around the persona of an agricultural policy analyst / farm planner, covering what they Think & Feel, Hear, See, and Say & Do, along with their Pain and Gain points.

Key insight: the core unmet need is a clear, visual way to compare existing crop production data across states, seasons, and years — not additional raw data collection (full canvas to be documented in the Ideation phase of the project repository).

3. Requirement Analysis

3.1 Customer Journey Map

The journey was mapped across stages — Discover, Explore, Compare, Decide, Reflect — tracing how a policymaker or farm planner moves from identifying a production question to confidently acting on state-wise or season-wise insights. Each stage was mapped to specific dashboard views (full swimlane map to be documented in the Requirement Analysis phase of the project repository).

3.2 Solution Requirement

Functional requirements cover data import and cleaning, state-wise production analysis, crop-wise analysis, season-wise analysis, yield analysis, production forecasting (machine learning), and dashboard/storyboard presentation. Non-functional requirements cover usability, security, reliability, performance, availability, and scalability (full FR/NFR tables to be documented in the Requirement Analysis phase of the project repository).

Requirement	Description
FR-1	Import and clean the historical crop production dataset (1997–2021)
FR-2	State-wise crop production analysis
FR-3	Crop-wise and season-wise production analysis
FR-4	Year-wise trend analysis
FR-5	Crop yield analysis and comparison
FR-6	Machine learning-based production forecasting
FR-7	Interactive dashboard and storyboard presentation

3.3 Data Flow Diagram

Data flows from the raw historical crop production dataset through Python-based cleaning and preprocessing (Pandas/NumPy), into a single Tableau data source, which feeds the dashboard worksheets; in parallel, the cleaned and feature-engineered data feeds a Scikit-learn regression model for production forecasting. Dashboard worksheets are assembled into a storyboard and published as a packaged workbook / repository deliverable (full DFD and user story table to be documented in the Requirement Analysis phase of the project repository).

Raw Dataset (CSV) → Python Cleaning & Preprocessing (Pandas, NumPy) → Tableau Data Source → Dashboard Worksheets + Storyboard | Feature Engineering → ML Model (Scikit-learn) → Forecast Output

3.4 Technology Stack

Category	Technologies
Programming Language	Python
Data Processing	Pandas, NumPy
Data Visualization	Tableau, Plotly, Matplotlib
Machine Learning	Scikit-learn
Dashboard	Tableau

4. Project Design

4.1 Problem Solution Fit

The Problem-Solution Fit canvas confirms the solution fits the validated problem: the Tableau workbook and forecasting model address the customer segment's job-to-be-done (compare and forecast crop production across states, crops, and seasons), remove the root cause (no aggregated visual or predictive view existed), and match observed stakeholder behavior (quick visual comparison and forward-looking planning instead of manual historical review) (full canvas to be documented in the Project Design phase of the project repository).

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard, supported by a Python/Scikit-learn forecasting model, built on the historical crop production dataset. Its novelty lies in unifying descriptive analysis (state, crop, season, year) with predictive analysis (production forecasting) in one consistent, filterable view, with direct value for regions and crops that show high year-to-year production volatility.

4.3 Solution Architecture

The architecture flows from the raw data layer (historical crop dataset) through a preparation layer (Python cleaning and feature engineering), into a single Tableau data source and an analysis layer (dashboard worksheets), a machine learning layer (production forecasting model), a presentation layer (storyboard), and a distribution layer (GitHub repository / packaged workbook) (full architecture diagram to be documented in the Project Design phase of the project repository).

5. Project Planning & Scheduling

Work was organized into 2 sprints covering the 7 functional requirements defined in the Requirement Analysis phase.

Sprint 1 — Data Foundation & Descriptive Analysis (10 story points)

- FR-1: Import & clean the 1997–2021 crop production dataset — 3 points
- FR-2: State-wise production analysis (map + bar charts) — 3 points
- FR-3: Crop-wise and season-wise production analysis — 2 points
- FR-4: Year-wise trend analysis — 2 points

Sprint 2 — Advanced Analysis, Forecasting & Presentation (10 story points)

- FR-5: Crop yield analysis and comparison — 3 points
- FR-6: ML-based production forecasting — 4 points
- FR-7: Interactive dashboard + storyboard assembly (3-scenario story) — 3 points

Total: 20 story points across 10 user stories. Both sprints were completed on schedule with an average velocity of 4 story points per day.

6. Functional and Performance Testing

6.1 Performance Testing

All records across the 1997–2021 dataset were verified to render correctly into the Tableau data source.

The following dashboard elements were validated:

- KPI summary card calculations: Avg. Yield (8,487.41), Area (M) (4,031.13), Production (M) (330,997.84)
- State-wise map rendering and colour-coded production shading
- Crop-wise and state-wise bar chart filters (Cultivated Area by Crop, State-wise Cultivated Area Comparison)
- Year-wise trend line (Avg. Yield 1997–2021) and year-wise production bar chart
- Season-wise donut chart (Whole Year, Rabi, Winter) and Production Distribution by Unit Type donut (Tonnes, Bales, Nuts)
- The 3-scenario storyboard (Farmer, Policy Analyst, Agribusiness Investor) navigation and panel transitions

All dashboard elements rendered and behaved as expected with no calculation errors observed.

7. Results

7.1 Output Screenshots

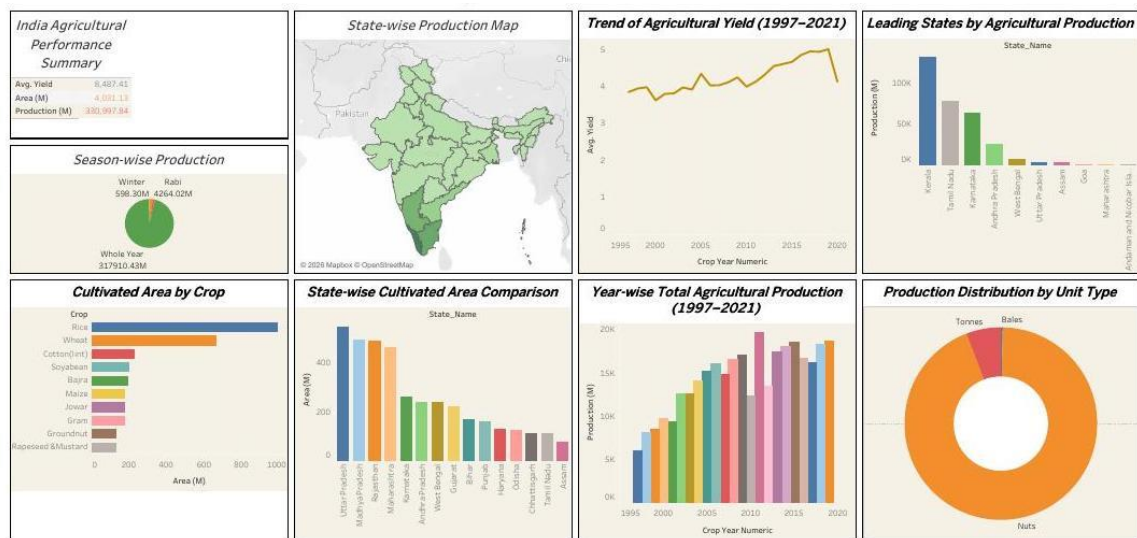


Figure 1: India Agricultural Performance Dashboard

The summary panel shows overall Production of 330,997.84M, Cultivated Area of 4,031.13M, and an Avg. Yield metric of 8,487.41 across the full 1997–2021 dataset.

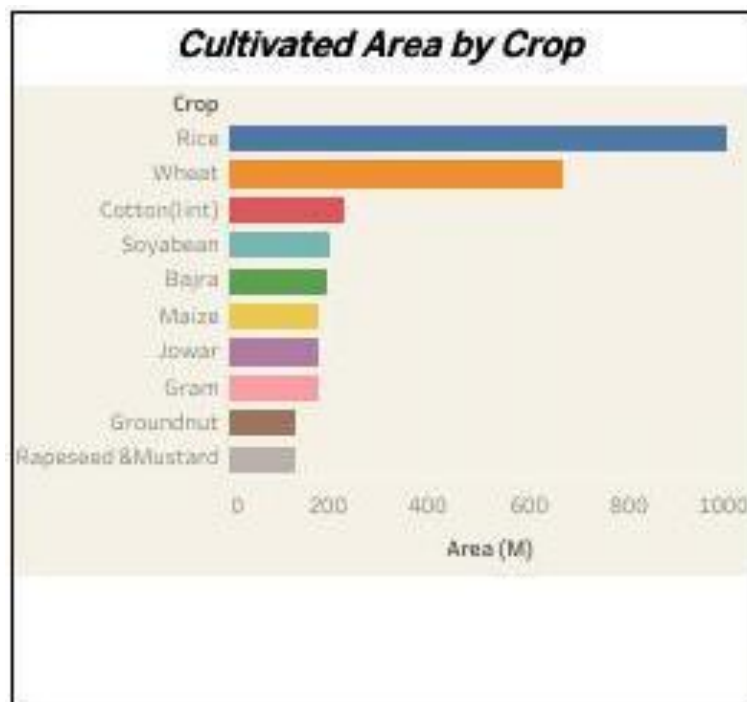


Figure 2: Cultivated Area by Crop

Rice occupies the largest cultivated area, followed by Wheat, Cotton (lint), Soyabean, Bajra, Maize, Jowar, Gram, Groundnut, and Rapeseed & Mustard.

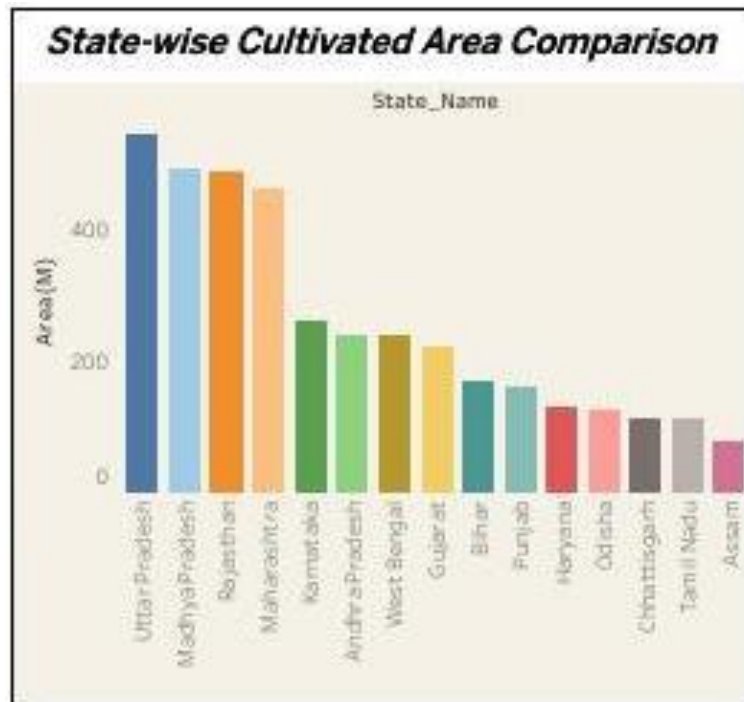


Figure 3: State-wise Cultivated Area Comparison

Uttar Pradesh leads in cultivated area, followed by Madhya Pradesh, Rajasthan, Maharashtra, and Karnataka — consistent with these states' traditional role as India's largest agrarian states.

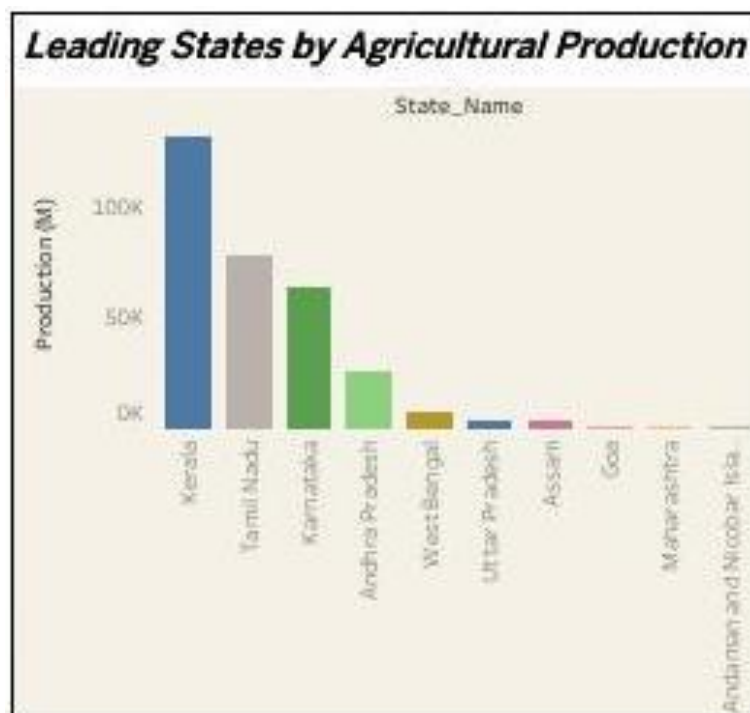


Figure 4: Leading States by Agricultural Production

Kerala tops the production ranking, followed by Tamil Nadu, Karnataka, Andhra Pradesh, and West Bengal. This ranking is influenced by the "Nuts" unit-type dominance in the dataset (coconut counted by nut count rather than tonnage), which inflates Kerala and other coconut-producing states relative to their cultivated area.

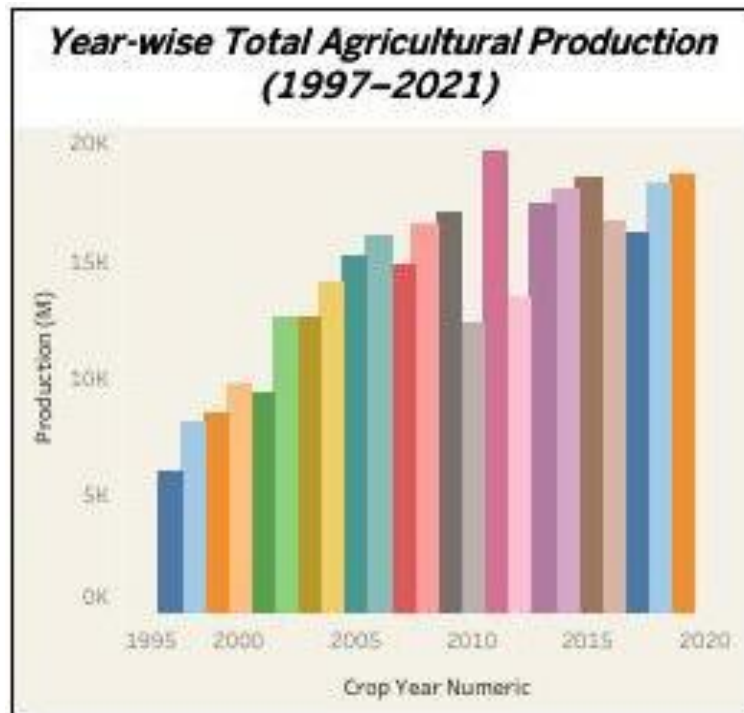


Figure 5: Year-wise Total Agricultural Production (1997-2021)

Production rose steadily from the late 1990s, accelerating through the 2000s and peaking in the mid-to-late 2010s, with a slight decline toward 2021.

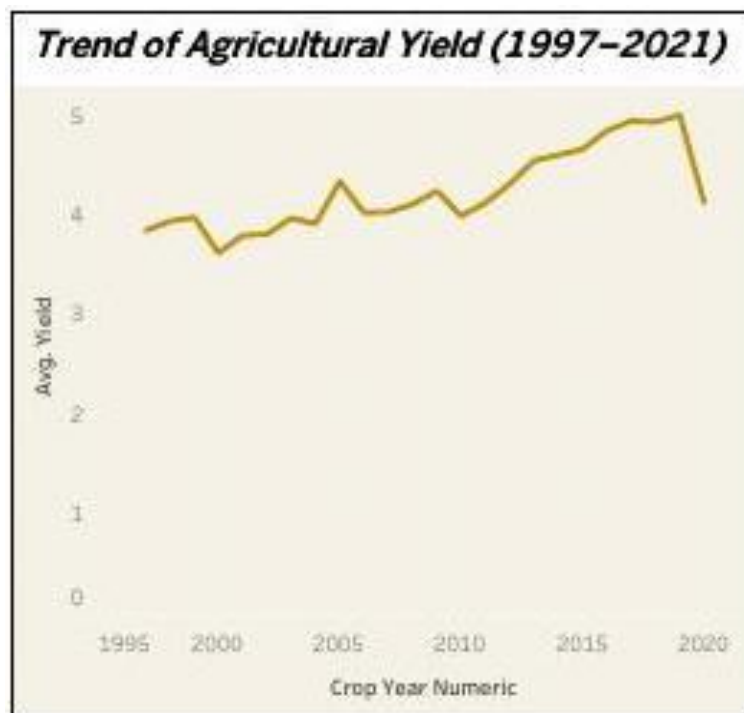


Figure 6: Trend of Agricultural Yield (1997-2021)

Average yield climbed gradually from around 3.5 to a peak near 5 around 2018-2020, followed by a small dip in the most recent year.



Figure 7: Season-wise Production

Whole Year cropping accounts for the overwhelming share of production (317,910.43M), with Rabi (4,264.02M) and Winter (598.30M) contributing much smaller shares.

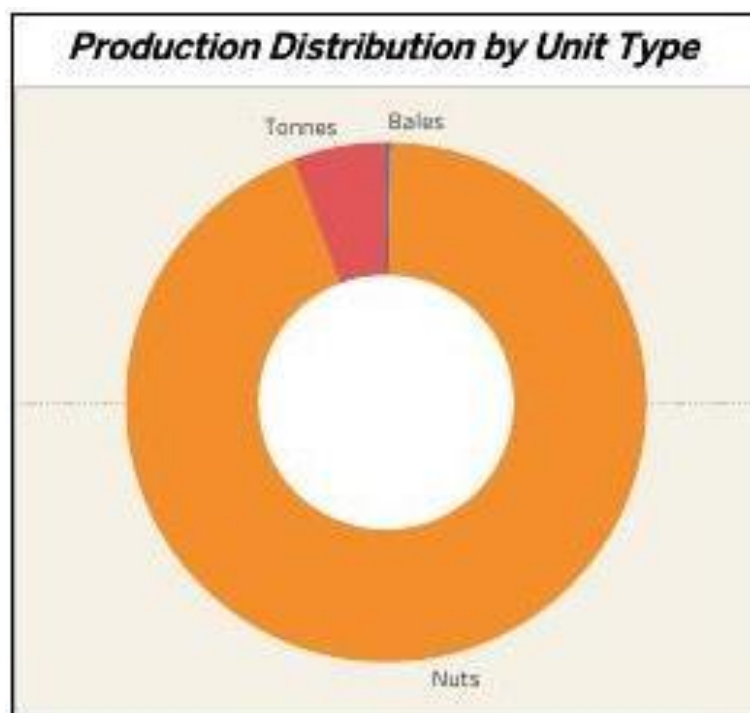


Figure 8: Production Distribution by Unit Type

Nuts dominate the unit-type breakdown, with Tonnes and Bales making up the remainder — reflecting the coconut-heavy composition of the underlying dataset.

7.2 Storyboard Scenarios

The storyboard presents the dashboard through three role-based scenarios:

- Scenario 1 — The Small-Scale Farmer: Rajesh, a farmer in Uttar Pradesh, uses seasonal and crop-level views to decide which crops to prioritize each season.
- Scenario 2 — The Government Policy Analyst: Ananya, from the Ministry of Agriculture, uses state-wise performance and long-term trend views to identify low-yield regions for resource allocation.
- Scenario 3 — The Agribusiness Investor: Vikram uses production growth and leading-state data to identify high-growth, lower-risk investment opportunities.

8. Advantages & Disadvantages

8.1 Advantages

- Consolidates decades of scattered agricultural records into one interactive, filterable workbook.
- Lets policymakers and analysts quickly identify high-performing states, crops, and seasons.
- Adds a forecasting layer that extends historical analysis into forward-looking planning.
- Single data source keeps all dashboard worksheets consistent and easy to maintain.

8.2 Disadvantages

- Static historical dataset (1997–2021) — does not reflect real-time production or weather conditions.
- No soil-quality or weather-variable integration despite these being major yield drivers.
- Forecasts are model-based estimates and should be validated against ground data before policy use.

9. Conclusion

This project transforms a raw, multi-decade agricultural dataset into an interactive, decision-ready Tableau workbook supported by a machine learning forecasting layer. The analysis is expected to confirm that crop production is unevenly distributed across states and seasons, concentrated in a handful of high-output states and staple crops, and shaped by monsoon-dependent seasonal cycles. These insights directly answer the questions raised during the Ideation and Requirement Analysis phases and demonstrate how visual analytics combined with predictive modeling can simplify decision-making for farmers, policymakers, and agribusiness analysts.

10. Future Scope

- Integrate weather and rainfall data to strengthen the link between climate variables and production.
- Incorporate satellite imagery for real-time crop health and area-under-cultivation monitoring.
- Add an AI-based crop recommendation layer tailored to region and season.
- Extend the forecasting model with real-time or annually updated data feeds.
- Publish the dashboard for broader public and policy-maker access.

11. Appendix

Dataset Link

Dataset: Historical Agricultural Crop Production Dataset (1997–2021), including crop production, cultivated area, season, state, district, and yield information — included in the project repository.

GitHub & Project Demo Link

GitHub Repository: github.com/ram-1709/Indian-s-Agriculture-Crop-Production-Analysis